optiDEM

The Only AI Integrated DEM Platform

# Product Brochure

AI Integrated Granular Flow Simulation Platform

**10M+**
Particles

**∞**
Cores

**AI**
Powered

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Chapter One

01

# Product Overview

The Future of Granular Flow Simulation

Revolutionary DEM Technology

Artificial Intelligence

Product Overview

# What is optiDEM?

## The Only AI-Integrated DEM Platform

**optiDEM** fuses cutting-edge algorithms with intelligent interaction, delivering a powerful, flexible, and highly customizable Discrete Element Method (DEM) solution for both research and industrial applications.

Unlike traditional DEM software that operates as a "black box," optiDEM provides unprecedented transparency and control. Our API architecture allows engineers and researchers to modify kernel algorithms, implement custom contact models, and extend functionality to meet their specific simulation requirements.

### AI-Powered

Natural language interface for script generation and parameter optimization

### Industry Templates

Standard templates covering many industies for quick setup and simulation

### Dual Mode

Intuitive GUI for beginners, powerful scripting for advanced users

### Unlimited Scale

No core limits or installation restrictions—scale from workstation to HPC

"optiDEM represents a paradigm shift in granular flow simulation—combining the power of scripting with AI-driven intelligence."

Industry Recognition

2024 Simulation Innovation Award

### Unique Market Position

- Only AI-integrated DEM platform
- dual mode operation
- Industry-specific templates
- Unlimited parallel computing

Performance Metrics

# Performance at Scale

**10M+**

**Particles Supported**

Simulate massive granular systems with millions of interacting particles

**∞**

**Compute Cores**

No artificial limits—scale from single workstation to HPC clusters

**100%**

**Industry Proof**

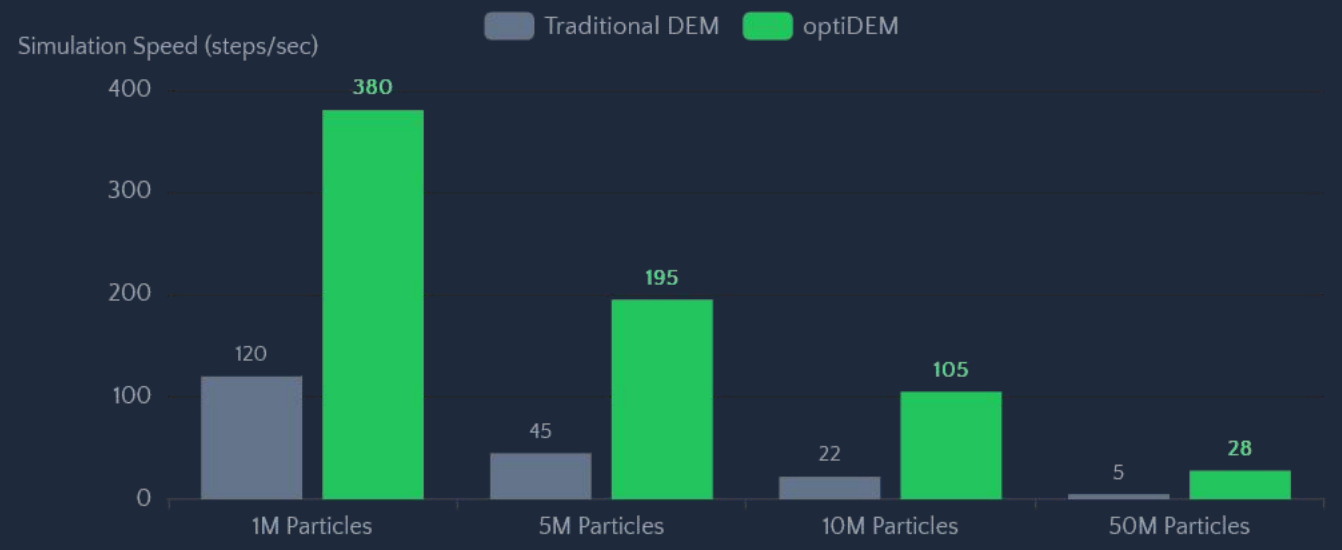
Validated by leading companies across mining, pharma, and chemical sectors

**AI**

**Integrated Assistant**

Natural language interface for instant script generation and optimization

## Computational Performance Comparison




## Proprietary Parallel Algorithms

Our custom-built parallel computing engine delivers **3-5x faster performance** compared to traditional DEM software, with linear scaling across thousands of cores.

Linear scaling efficiency up to 10,000+ cores

## Hardware Flexibility

Run on any hardware configuration—from laptops for prototyping to **GPU-accelerated workstations** and distributed HPC clusters.

Workstation | Server | HPC Cluster

Chapter Two

02

# Core Advantages

Why optiDEM Stands Apart

Breaking Technical Barriers

Core Advantage 01

# AI-assisted Architecture

## Industry's Only DEM Platform with AI Access

**optiDEM breaks the mold** by being the only DEM platform that offers complete AI integration. This revolutionary approach empowers users to **modify and extend kernel algorithms freely** , achieving true personalized simulation customization.

Traditional DEM software operates as a "black box," limiting researchers and engineers to predefined models and parameters. With optiDEM, you have complete transparency and control over every aspect of the simulation engine.

### Full Transparency

Examine, understand, and validate every algorithm and calculation in the simulation engine

### Custom Extensions

Implement unique contact models, coupling algorithms, and material behaviors

### Research Freedom

Publish and share modifications without licensing restrictions

### Community Driven

Benefit from and contribute to a growing ecosystem of extensions

### Kernel Architecture

```
// Custom Contact Model Example
class CustomContact :
  ContactModel :
  def calculate_force (self):
    # Your custom physics
    force = self .stiffness *
      overlap * self .damping
    return force
```

Easy to extend with Python/C++

### Use Cases

- Novel material models for composite particles
- Multi-physics coupling (CFD-DEM, FEM-DEM)
- Custom breakage and wear models
- Specialized boundary conditions

Core Advantage 02

# AI Intelligent Assistant

## Natural Language Interface for Simulation

**The integrated AI chat module** transforms how users interact with DEM simulations. Instead of manually writing complex scripts or navigating through menus, simply **describe what you want in natural language** and the AI assistant generates the appropriate configuration, scripts, and parameters.

This breakthrough capability dramatically reduces the learning curve for new users while accelerating workflow for experienced engineers. The AI understands domain-specific terminology and can provide intelligent optimization recommendations based on your simulation goals.

### Script Generation
Convert natural language to executable code

### Smart Recommendations
Get optimization suggestions automatically

### Instant Q&A
Query parameters and get explanations

### Productivity Impact
Users report **60% reduction** in setup time and **40% faster** optimization cycles

## AI Assistant Console Demo

AI Assistant Console

**User:** Generate a script for pharmaceutical mixing simulation with 50k particles

**AI:**

Analyzing request...

✓ Detected: Pharma mixing application

✓ Particle count: 50,000

✓ Loading template: Pharma_Mix_v2.cfg

Generated Python script:

```
import
  optidem # Load template
  sim = optidem.load_template( 'pharma_mix' )
  # Configure particles
  sim.particles.count = 50000
  # Run simulation
  sim.run(device= 'cuda' )
```

Script ready! Estimated runtime: 12 minutes

**User:** What contact model should I use for cohesive powders?

**AI:**

For cohesive pharmaceutical powders, I recommend:

→ Johnson-Kendall-Roberts (JKR) model

→ Cohesion energy: 0.5-2.0 J/m$^2$

→ Surface energy calibration recommended

Would you like me to apply these settings?

Core Advantage 03

# Dual Mode Operation & Unlimited Computing

## Dual Mode Operation

**optiDEM balances convenience with flexibility** through its innovative dual-mode architecture. Beginners can leverage the intuitive GUI with pre-set templates, while experts can harness powerful scripting for complex workflows and batch processing.

### GUI Mode

- Drag-and-drop geometry setup
- Visual particle generation
- Real-time simulation monitoring
- Pre-configured industry templates

Ideal For

New users, quick prototyping, standard simulations

### Script Mode

- Parametric studies & optimization
- Batch processing workflows
- Custom algorithm implementation
- Integration with external tools

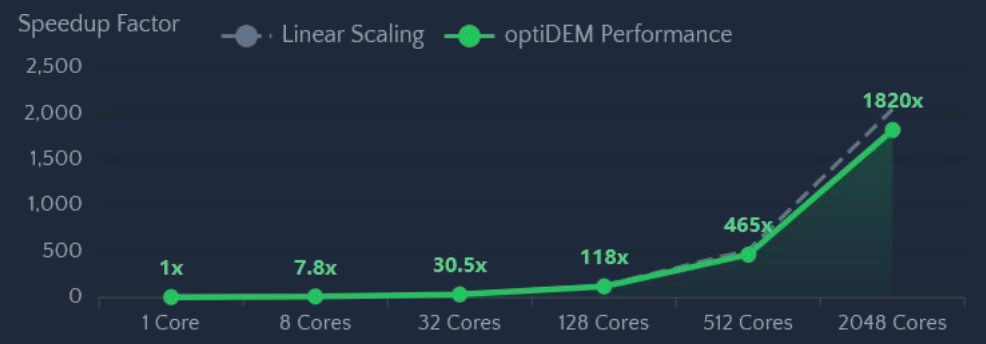
Ideal For

Advanced users, research, complex automation

## Unlimited Parallel Computing

**No artificial limits.** optiDEM's proprietary parallel algorithms come with no limits on core count or installations . Whether you're running on a single workstation or a thousand-node HPC cluster, optiDEM scales seamlessly to meet your computational needs.

### Scaling Performance




| ∞ | ∞ | GPU |
|---|---|---|
| No Core Limits | No Install Limits | CUDA Support |

Chapter Three

03

# Industry Applications

Applied Across Industries

From Research to Production

Industry Solutions

# Industry Templates & Use Cases

## Mining

Optimize mineral processing operations with validated simulation models

- Crushing & grinding
- Screening & classification
- Conveying systems
- Stockpiling & reclaiming

## Pharmaceutical

Ensure product quality with precise powder handling simulations

- Powder mixing & blending
- Granulation processes
- Tablet coating
- Filling & packaging

## Chemical

Design efficient reactors and processing equipment

- Drying operations
- Fluidized bed reactors
- Pneumatic conveying
- Catalyst handling

## Agriculture

Optimize seed and feed processing operations

- Seed processing
- Feed production
- Fertilizer handling
- Grain storage

## Energy

Model combustion and energy storage systems

- Coal combustion
- Biomass processing
- Particle energy storage
- Catalyst systems

### Pre-Configured Industry Templates

Rapidly build industrial models without starting from scratch

**50+** Templates

**100%** Validated

**5min** Setup Time

Industry Validation

# Trusted by Industry Leaders

## Proven Across Multiple Sectors

**Leading companies worldwide** rely on optiDEM for their critical simulation needs. From global mining giants to pharmaceutical innovators, our platform has been validated in the most demanding industrial environments.

This diverse portfolio of industry leaders demonstrates optiDEM's **versatility, reliability, and proven performance** across different applications and scales.

| | | | | |
|---|---|---|---|---|
| **BHP** Mining | **Rio Tinto** Mining | **Chinalco** Mining | **Zijin** Mining | **Bisalloy** Materials |
| **Amman** Pharma | **CPM** Equipment | **Famsun** Agriculture | **MRL** Research | **NStar** Chemical |

**10+** Industry Partners

**5** Major Sectors

**100%** Validation Rate

## Client Testimonials

"optiDEM's open-source approach allowed us to implement a custom breakage model that improved our mill optimization by 15%."

**Dr. Sarah Chen**
Principal Engineer, BHP

"The AI assistant cut our simulation setup time in half. What used to take days now takes hours."

**Marcus Rodriguez**
Simulation Lead, Amman

"We scaled from 8 cores to 512 cores seamlessly. The performance scaling is truly impressive."

**Prof. Li Wei**
Director, MRL

Workflow

# Technology Workflow

## Flexible Three-Step Process

**optiDEM balances convenience with flexibility** for users at all levels. Whether you're a beginner using the GUI or an expert leveraging scripting, our workflow adapts to your needs while maintaining full power and control.

### 1 Visual Setup

Use the intuitive GUI for drag-and-drop geometry setup, visual particle generation, and real-time simulation monitoring. Perfect for rapid prototyping and standard simulations.

Drag & Drop | Real-time Preview | Templates

### 2 AI Script Generation

Leverage natural language commands to generate complex simulation scripts instantly. The AI understands domain terminology and provides intelligent optimization recommendations.

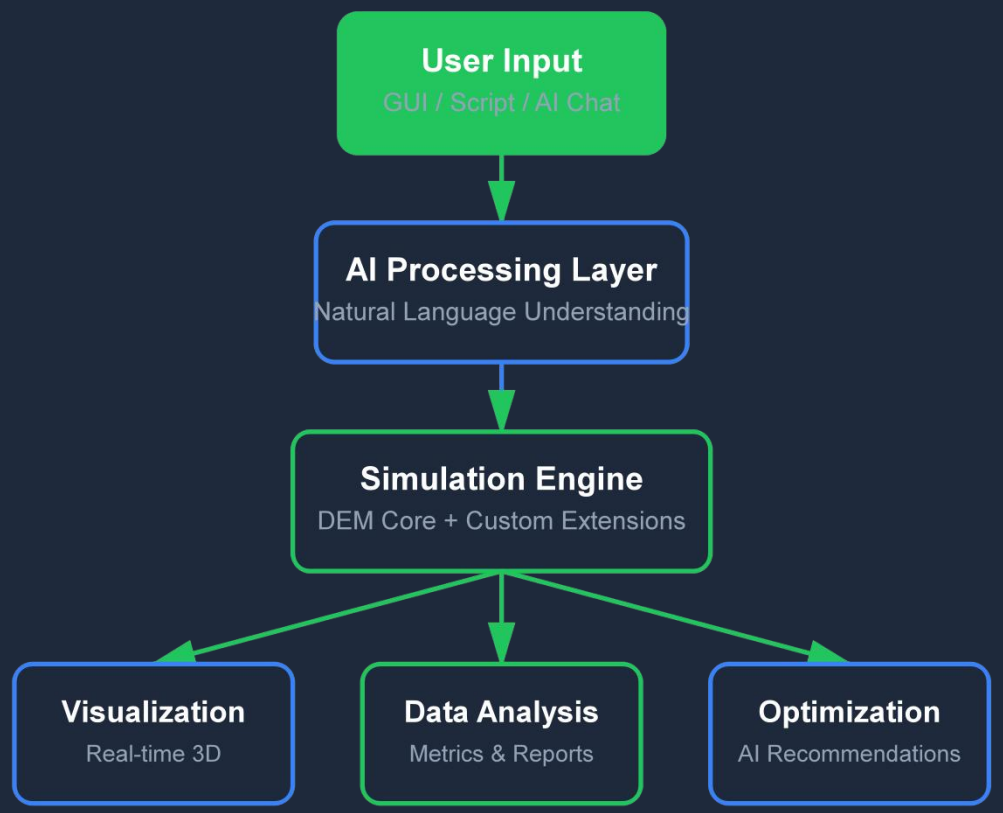
Natural Language | Auto-Optimization | Batch Processing

### 3 Custom Kernel Extension

Access the source code to implement unique contact models, coupling algorithms, or specialized boundary conditions. Full transparency for advanced research and development.

Source Access | Custom Physics | Multi-physics

## Workflow Architecture




| 3x | 60% | 100% |
|---|---|---|
| Faster Setup | Time Savings | Flexibility |

Get Started Today

# Ready to Transform Your Simulation Workflow?

Join the open source revolution in granular flow simulation.

No core limits. No black boxes. Full transparency.

Download Brochure

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Real-world applications

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Expert assistance

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optiDEM
Precision Granular Flow Simulation

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